

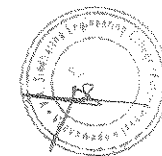


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VELAGAPUDI RAMAKRISHNA
SIDDHARTHA ENGINEERING COLLEGE
(AUTONOMOUS)



IV/IV B.Tech. DEGREE EXAMINATION, NOVEMBER - 2023

Seventh Semester

INFORMATION TECHNOLOGY

20IT7301 DEEP LEARNING

Time: 3 hours

Max. Marks: 70

Part-A is compulsory

Answer One Question from each Unit of Part - B

Answer to any single question or its part shall be written at one place only

PART-A

10 x 1 = 10M

1.
 - a. What is auto-association task in neural networks? (CO1 K1)
 - b. Write the mathematical formula for summation function. (CO1 K1)
 - c. Define over fitting. (CO1 K1)
 - d. What is feature set? Give one example. (CO2 K1)
 - e. Render simple architecture for auto encoder. (CO2 K1)
 - f. Define max pooling. (CO2 K1)
 - g. Render model for basic RNN. (CO3 K1)
 - h. Specify layers in deep CNN (CO3 K1)
 - i. Give two applications for Generative adversarial neural network. (CO4 K1)
 - j. Define conditional Generative adversarial neural network. (CO4 K1)



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PART-B

4 x 15 = 60M

UNIT-I

2. a. Describe architecture of feed forward neural networks. **(CO1 K2) 8M**
b. Discuss the importance of training, validation and testing sets in modeling neural networks. **(CO1 K3) 7M**
(or)
3. a. Define linear neuron. List limitations of linear functions. **(CO1 K2) 7M**
b. Explain and analyze delta rule with example. **(CO1 K3) 8M**

UNIT-II

4. a. Explain about CNN architecture. **(CO2 K3) 8M**
b. Elaborate on CNN for NLP and Time Series. **(CO2 K3) 7M**
(or)
5. a. Explain about sparsity in auto encoders. **(CO2 K3) 7M**
b. Describe how effective dimensionality reduction can be done using auto encoders. **(CO2 K3) 8M**

UNIT-III

6. a. Elaborate on sequence to sequence architectures in auto encoders. **(CO3 K3) 7M**
b. Discuss on computational graphs using in sequence modeling. **(CO2 K3) 8M**

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(or)



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7. a. Describe about deep RNN for identifying hand written digits as case study. **(CO3 K3) 8M**
b. Explain about Forward Propagation in Deep Networks and mathematical notations of hyper parameters. **(CO3 K3) 7M**

UNIT-IV

8. a. Discuss on various attention mechanisms applicable for recurrent models. **(CO4 K3) 8M**
b. Illustrate steps required to generate image data using generative adversarial networks. **(CO4 K4) 7M**
(or)

9. a. Machine translation is challenging task in implementing real time use case. Justify on this statement with suitable examples. **(CO4 K4) 8M**
b. List out various challenges of deep learning and neural networks. **(CO4 K3) 7M**

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